Photograph Mexy Xavier Imaging Atul Deshmukh

Say good-bye to endless cables and entangled wires, and say hello to a whole new workplace

Break Free

Mustali Kachwala

Wi-Fi, or Wireless Fidelity, means freedom—freedom to connect to the Internet from just about anywhere in your home or small office. We are all familiar with the cell-based technique of mobile communication. It is the basis of Wi-Fi technology too. As long as you are in a region covered by Wi-Fi—also called a hotspot—you can access the Internet.

While on one hand, Wi-Fi gets rid of problems, such as wire hassles, that exist with wired networking, it does bring about a setback in the data throughput provided by the various Wi-Fi standards, *vis-à-vis* wired networking. Here, we check which products in this domain are the best. How large is the bandwidth gap between wired and wireless? How much extra do you pay? We explore several such questions and attempt to provide answers.

SETTING UP A NEW OFFICE

Site Survey

A major hurdle you encounter while setting up a Wi-Fi hotspot is determining the exact location for Access Points (APs), and the number of APs you need to provide a constant and reliable connection. Install too many devices and you end up paying more than needed. This also increases the chances of RF interference with nearby APs.

In its simplest form, you can conduct a site survey with just a Wi-Fi enabled laptop and appropriate software. A comprehensive site survey can be conducted with devices such as spectrum or handled hardware analyser from, say, AirMagnet, especially designed for this purpose. However, these are expensive and you might want to take the laptop route.

The site survey process consists of first checking whether any device is causing interference in the region that you need covered. You might find signals from neighbouring areas. If you do find interference, locate that is causing it and reposition the equipment.

The next step involves placing each AP at different locations, one at a time, and marking out the boundaries where the signal is available. The entire process is based on trial and error. Once you prepare a boundary map by placing an AP at multiple locations, analyse it to ensure a proper overlap of signals, and make sure that